

Exploratory Data Analysis (EDA) Project

Retail Sales & Customer Analysis — Student Project

Dataset: EDA_Retail_Sales_Project_Dataset.xlsx | 1,020 raw rows including 20 intentional duplicates | 17 columns

1. Data Understanding

1. Load the Raw_Data sheet.
2. Display head(), tail(), shape, columns, dtypes and info().
3. Generate descriptive statistics for numerical and categorical columns.
4. Identify numerical, categorical, date and identifier columns.

2. Missing Values

1. Find missing values per column.
2. Display only columns containing missing values.
3. Calculate missing-value percentages.
4. Choose and justify suitable treatments for missing City, CustomerAge, CustomerRating, PaymentMode and CustomerSegment.

3. Duplicates

1. Find the number of duplicate rows.
2. Display duplicate records.
3. Remove exact duplicates from a working copy.
4. Verify the final row count.

4. Derived Columns

1. Create GrossAmount = Quantity × UnitPrice.
2. Create DiscountAmount = GrossAmount × DiscountPct / 100.
3. Create NetSales = GrossAmount – DiscountAmount.
4. Use apply() to classify NetSales as High Value (> ₹50,000) or Regular.
5. Create OrderMonth from OrderDate.

5. Selection & Filtering

1. Select key sales columns.
2. Filter NetSales > ₹20,000.
3. Filter Electronics orders.
4. Filter Hyderabad orders with NetSales > ₹15,000.
5. Filter Quantity between 3 and 5.
6. Filter UPI or Credit Card orders.
7. Filter returned orders.
8. Filter customers younger than 25 or older than 50.

6. Sorting & Frequency Analysis

1. Sort by NetSales descending.
2. Find unique categories and number of unique products.
3. Use value_counts() for PaymentMode, SalesChannel, Product and State.
4. Calculate sum, mean, min and max of NetSales.
5. Find the top 5 products and top 5 cities by NetSales.

7. GroupBy & Aggregation

1. Group by Category: total sales, average sales and total quantity.
2. Group by State: total NetSales.
3. Group by SalesChannel and CustomerSegment: total NetSales.
4. Identify the best-performing category, state, channel and segment.

8. NumPy Analysis

1. Convert NetSales to a NumPy array.
2. Calculate mean, median, standard deviation, minimum and maximum.
3. Use argmax() and argmin().
4. Use Boolean filtering for values above the overall average.
5. Use np.where() to classify values.

9. Matplotlib

1. Bar chart: orders by Category.
2. Horizontal bar chart: top 10 products by NetSales.
3. Line chart: monthly NetSales.
4. Histogram: NetSales distribution.
5. Scatter: Quantity vs NetSales.
6. Box plot: NetSales distribution.
7. Pie chart: SalesChannel share.
8. Bar chart: average NetSales by CustomerSegment.
9. Add data labels to a suitable bar chart and annotate the highest monthly value.

10. Seaborn

1. barplot: average NetSales by Category.
2. barplot with hue='Gender'.
3. scatterplot: Quantity vs NetSales with hue=CustomerSegment and style=SalesChannel.
4. lineplot: monthly NetSales with marker='o' and errorbar=None.
5. boxplot: NetSales by Category with hue=CustomerSegment.
6. histplot: NetSales with bins and KDE.
7. countplot: orders by SalesChannel with hue=CustomerSegment.
8. heatmap: correlations among suitable numerical columns with annot=True.

11. Business Questions

1. Which category has the highest NetSales?
2. Which state performs best?
3. Which sales channel performs best?
4. Which customer segment has the highest average order value?
5. Does Quantity appear related to NetSales?
6. Which products show unusual values or possible outliers?
7. What is the return rate?
8. Which month has the highest NetSales?
9. Write at least 5 actionable business insights.

12. Final Deliverables

1. Completed EDA notebook.
2. Cleaned DataFrame.
3. At least 10 meaningful charts.
4. Short findings summary.
5. At least 5 business recommendations.

Important: Every important visualization must be followed by a short interpretation explaining what it shows and why it matters.